GENERATORS WHOSE SQUARE AND CUBE LIE IN THE BASE, WITH LOCALLY NILPOTENT KERNEL, GIVE A UNIVERSAL HOMEOMORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Closedness of the map on spectra for integral ring maps. Invoked in the counted proof at lines 11000-11002. Gives closedness of the induced map on spectra.
- Density of the image from generic points. Invoked in the counted proof at lines 11002-11003 and again at the end. Supplies density of the image, which together with closedness gives surjectivity.

Source reference. Original lemma 00HZ: sources/stacks-algebra/pinned-original-source.tex, source lines 9869–9869. Navigation pointer for \ref{lemma-going-up-closed} used inside the counted span (line 11001); the argument that consumes it is retained in the span.

Source reference. Original lemma 00FL: sources/stacks-algebra/pinned-original-source.tex, source lines 6138–6138. Navigation pointer for \ref{lemma-image-dense-generic-points} used inside the counted span (line 11003); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00GU: sources/stacks-algebra/pinned-original-source.tex, source lines 8070–8070. Navigation pointer for \ref{lemma-integral-going-up} used inside the counted span (line 11002); the argument that consumes it is retained in the span.

Source reference. Original lemma 00FI: sources/stacks-algebra/pinned-original-source.tex, source lines 6039–6039. Navigation pointer for \ref{lemma-surjective-spec-radical-ideal} used inside the counted span (line 11022); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $\varphi : R \to S$ be a ring map. Assume*

(a) *S is generated as an R-algebra by elements x such that $x^2, x^3 \in \varphi(R)$, and*
(b) *$\mathrm{Ker}(\varphi)$ is locally nilpotent,*

Then φ induces isomorphisms on residue fields and a homeomorphism of spectra. For any ring map $R \to R'$ the ring map $R' \to R' \otimes_R S$ also satisfies (a) and (b).

Proof. Assume (a) and (b). The map on spectra is closed as S is integral over R, see Lemmas 00HZ and 00GU. The image is dense by Lemma 00FL. Thus $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is surjective. If $\mathfrak{q} \subset S$ is a prime lying over $\mathfrak{p} \subset R$ then the field extension $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is generated by elements $\alpha \in \kappa(\mathfrak{q})$ whose square and cube are in $\kappa(\mathfrak{p})$. Thus clearly $\alpha \in \kappa(\mathfrak{p})$ and we find that $\kappa(\mathfrak{q}) = \kappa(\mathfrak{p})$. If $\mathfrak{q}, \mathfrak{q}'$ were two distinct primes lying over $\mathfrak{p}$, then at least one of the generators x of S as in (a) would have distinct images in $\kappa(\mathfrak{q}) = \kappa(\mathfrak{p})$ and $\kappa(\mathfrak{q}') = \kappa(\mathfrak{p})$. This would contradict the fact that both x^2 and x^3 do have the same image. This proves that $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is injective hence a homeomorphism (by what was already shown).

Since φ induces a homeomorphism on spectra, it is in particular surjective on spectra which is a property preserved under any base change, see Lemma 00FI. Therefore for any $R \to R'$ the kernel of the ring map $R' \to R' \otimes_R S$ consists of nilpotent elements, see Lemma 00FL, in other words (b) holds for $R' \to R' \otimes_R S$. It is clear that (a) is preserved under base change. □